

DAY-01 – CYBER SECURITY

Clean and organized study notes based on the provided notebook

1. Cyber Security Basics

- Cybersecurity protects digital assets, systems, networks, applications, and physical resources from unauthorized access, damage, disruption, or misuse.
- Information security focuses on protecting information in both physical and digital forms.
- Hacking refers to gaining access to a system by identifying and exploiting weaknesses or vulnerabilities. Ethical hacking is performed with authorization.

2. Operating Systems

- Client/End-user OS: used directly by people on computers and other end-user devices. Examples include Windows 7 and Windows 10.
- Server OS: designed to provide services over a network, such as file, web, database, authentication, or application services. Examples include Windows Server 2012, 2016, 2019, 2022, and 2025.

3. Types of Servers

- Web server – delivers web pages and web resources.
- Web application server – runs application/business logic.
- Database server – stores and manages databases.
- E-mail server – handles sending, receiving, and storing email.
- Gaming server – supports multiplayer game sessions.
- Proxy server – acts as an intermediary between clients and other servers.
- Reverse proxy – sits in front of servers and forwards client requests to the appropriate backend.
- Network server – provides network-related services.
- File server – provides centralized file storage and access.

4. Types / Areas of Cybersecurity

- Physical security, perimeter security, application security, identity and access management (IAM), cloud security, data security, email security, AI security, operating-system security, hardware security, firmware security, endpoint security, container security, and network security.

5. OSI Model and Networking

- The OSI (Open Systems Interconnection) model describes network communication using seven layers.
- 1. Application – provides network services to applications.
- 2. Presentation – handles data representation, translation, encryption, and compression.
- 3. Session – establishes and manages communication sessions.

- 4. Transport – provides end-to-end transport and reliability functions.
- 5. Network – handles logical addressing and routing.
- 6. Data Link – handles frames, MAC addressing, and local delivery.
- 7. Physical – transmits bits over physical media.

6. Digital Data and Security

- Digital data is information stored or processed electronically. Data security protects it against unauthorized access, alteration, disclosure, loss, or destruction.
- Cloud storage and services such as Amazon S3 are examples of places where digital data can be stored.
- Governance and compliance help organizations manage security according to policies, standards, laws, and regulations.
- Industrial security protects operational/industrial systems. Examples mentioned in the notes include PLC and SCADA environments.

7. Core Cybersecurity Topics

- Important foundational areas include operating-system security, networking, file servers, viruses and antivirus concepts, forensics, data leakage, denial-of-service (DoS), malware, web servers, load balancing, and network security.

8. IP Addressing

- IPv4 uses 32-bit addresses. IPv6 uses 128-bit addresses.
- Traditional IPv4 classes are: Class A (1–126), Class B (128–191), Class C (192–223), Class D (224–239), and Class E (240–255).
- Private IPv4 ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.
- Private IP addresses are used inside local/private networks and are not directly routable on the public Internet. Public IP addresses are used for Internet-facing communication.
- CIDR (Classless Inter-Domain Routing) is a method for representing IP networks using a prefix length, such as /24.

9. Subnetting, Routing and Protocols

- Subnetting divides a network into smaller logical networks (subnets).
- Supernetting combines multiple smaller networks into a larger address block.
- Routing is the process of selecting paths for packets between networks.
- Common networking protocols include TCP/IP, UDP, IPv4, IPv6, ICMP, DHCP, and DNS.
- RIP (Routing Information Protocol) is a routing protocol; RIP version 1 and RIP version 2 are versions of RIP.

10. Network Topologies

- Bus topology – devices share a common backbone.
- Ring topology – devices form a circular path.
- Star topology – devices connect to a central switch or hub.
- Mesh topology – devices have multiple interconnections for redundancy.
- Tree topology – multiple star networks are arranged hierarchically.
- Hybrid topology – combines two or more different topologies.

11. Firewall and Network Protection

- A firewall controls and filters network traffic according to security rules.
- Common firewall approaches include packet filtering, stateful inspection, proxy firewalls, and next-generation firewalls (NGFW).
- Firewalls are an important component of network security, but they are only one layer of a complete security architecture.

Source basis: the six notebook images provided in this conversation. The clean version reorganizes the same topics and uses standard terminology where the intended meaning is clear.